Architectural Foundations of Event Density

Abstract

Event Density (ED) exhibits a reproducible architectural organization: saddles, channels, boundaries, and horizons emerge across a wide range of conditions and govern the formation, persistence, and interaction of individuating structures. This paper establishes the theoretical foundations underlying that architecture. We define the core ontological objects of ED and derive their mathematical origins in gradient sign-flip geometry, curvature transitions, and determinant-zero surfaces of the Hessian. We show that these motifs organize into distinct dynamical regimes—hyper-coherent, stable-coherent, marginal, and non-individuating—each characterized by specific configurations of curvature, noise, and mobility. Through analysis of parameter deformation, noise thresholds, and mobility laws, we demonstrate that the architectural motifs are universal: they arise independently of initial conditions and persist across broad parameter ranges. Extending the framework to higher dimensions reveals that the same structural logic generalizes naturally, with saddles becoming sheets, channels becoming tubes, and boundaries and horizons becoming surfaces. Together, these results establish ED as a coherent architectural framework in which structure is not imposed but emerges inevitably from the dynamics of the event density field.

Introduction

The Event Density framework reveals a striking and repeatable architecture: saddles, channels, boundaries, and horizons emerge across a wide range of conditions, forming coherent structures that persist, interact, and organize the dynamics of the field. The first paper demonstrated this architecture empirically through controlled simulations. The present paper establishes the theoretical foundations that make this architecture inevitable.

At the core of ED is a simple principle: local gradients and curvature reorganize event density into stable geometric motifs. These motifs are not artifacts of initial conditions or parameter choices; they arise from the intrinsic structure of the ED rule. Gradient sign-flips generate saddles, curvature transitions generate boundaries and horizons, and coherent gradient alignment generates channels. Together, these motifs form the minimal ontology required to describe individuating structure in ED.

The goal of this paper is to formalize that ontology, derive its mathematical basis, and show that the resulting architecture is universal. We develop the geometric and dynamical conditions under which saddles, channels, boundaries, and horizons must appear, classify the regimes that govern their stability, and demonstrate that these structures persist under parameter deformation, noise, and dimensional uplift. In doing so, we establish ED as a general theory of architectural emergence: a framework in which structure is not imposed but arises from the dynamics themselves.

Ontology of Event Density Architecture

Event Density Field

The Event Density field $E(x,t)$ is the fundamental quantity of the framework. It assigns to every point in space–time a scalar value representing the local accumulation, suppression, or redistribution of events. All architectural structures arise from the geometry of E, its gradients, and its curvature.

Gradient Field

The gradient ∇E encodes the directional tendencies of the system. Its magnitude reflects the strength of local

change, and its orientation determines the flow of individuating structure. Architectural motifs emerge where the gradient changes sign, direction, or stability.

Hessian and Curvature
The Hessian $H = \nabla^2 E$ captures the local curvature of the field. Its eigenvalues describe compression and expansion along principal directions. Curvature transitions—especially where $\det(H) = 0$ —mark the boundaries between architectural regions and determine where saddles, channels, and horizons form.

Architectural Objects

Saddles
A saddle is the fundamental geometric unit of ED architecture. It is defined by a sign-flip of the gradient across orthogonal directions, producing one axis of compression and one axis of expansion. The saddle organizes the surrounding field into four sectors: two convergent, two divergent. Boundaries and horizons emanate from these sectors, making the saddle the anchor of higher-order structure.

Channels
A channel is a persistent corridor of aligned gradient flow. It forms when curvature organizes into a narrow region where ∇E maintains coherent direction over extended distance or time. Channels transmit individuating structure, connect saddles, and stabilize coherent regimes. Their width, persistence, and mobility depend on the mobility exponent and noise amplitude.

Boundaries
A boundary is a compression-dominated region where gradient vectors converge. Boundaries form the "walls" of ED architecture, separating coherent interiors from exterior fluctuations. They arise where curvature becomes positive along the transverse direction and where $\det(H)$ changes sign. Boundaries stabilize pockets, wells, and long-lived structures.

Horizons
A horizon is an expansion-dominated region where gradient vectors diverge. Horizons form the "open surfaces" of ED architecture, marking transitions from structured interiors to unstructured exteriors. They arise where curvature becomes negative along the transverse direction and where $\det(H)$ changes sign. Horizons regulate the flow of noise and determine whether structures persist or dissolve.

Regimes
A regime is a global organizational state of the ED field. Regimes are defined by the stability, persistence, and coherence of saddles, channels, boundaries, and horizons. The four canonical regimes—hyper-coherent, stable-coherent, marginal, and non-individuating—represent distinct phases of architectural behavior. Each regime corresponds to characteristic patterns of curvature, noise, and mobility.

Invariants
An invariant is a structural feature that persists under deformation of parameters, initial conditions, or noise. Invariants include the existence of saddles, the organization of channels, the presence of boundaries and horizons, and the qualitative transitions between regimes. Invariants define the architectural identity of ED and justify its universality.

Architectural Relations

Saddle–Channel Relation
Channels originate from saddles and often terminate at other saddles, forming networks of coherent flow. This relation is the backbone of ED structure.

Boundary–Horizon Complementarity
Boundaries and horizons arise as complementary surfaces around saddles and channels. Their interplay determines whether structures remain stable, propagate, or dissolve.

Regime–Invariant Coupling
Regimes are distinguished by which invariants remain stable under perturbation. Hyper-coherent regimes preserve all architectural invariants; marginal regimes preserve only some; non-individuating regimes preserve none.

Ontological Commitments
The ontology commits the ED framework to a structural view of dynamics: the system is defined not by its components but by the relations, transitions, and geometric motifs that emerge from the event density field. Saddles, channels, boundaries, horizons, and regimes are not optional features—they are the necessary architectural consequences of the ED rule.

Mathematical Structure of Event Density Architecture

Event Density Dynamics
The Event Density field $E(x,t)$ evolves according to a local update rule that redistributes density based on gradients, curvature, mobility, and noise. The general form is:

$$\partial E / \partial t = M(E, \nabla E) + C(H) + \eta(x,t) - S(E)$$

where:

- M encodes mobility-driven flow,
- C encodes curvature-driven reorganization,
- η is noise,
- S is suppression.

The specific form is less important than the structural consequences: gradients sharpen, curvature organizes, and noise perturbs but does not erase the architecture unless it exceeds a critical threshold.

Gradient Sign-Flip and the Genesis of Saddles

Orthogonal Sign-Flip Condition
A saddle forms where the gradient reverses sign across orthogonal directions:

$$\partial_x E > 0, \qquad \partial_y E < 0$$

in one quadrant, and the opposite in the adjacent quadrant.

This condition guarantees one axis of compression and one axis of expansion. It is the minimal geometric requirement for individuating structure.

Structural Consequences

- Two convergent sectors (boundary-forming).
- Two divergent sectors (horizon-forming).
- A natural alignment axis for channels.

This is the first point where the architecture becomes mathematically inevitable: any field with sufficient variation will produce sign-flip loci, and ED dynamics amplify them into saddles.

Curvature and the Role of the Hessian

Hessian Structure
The Hessian

$$H = \begin{pmatrix} \partial_{xx}E & \partial_{xy}E \\ \partial_{yx}E & \partial_{yy}E \end{pmatrix}$$

encodes local curvature. Its eigenvalues λ1,λ2 determine whether the region is compressive, expansive, or mixed.

det(H) = 0 Curves
The curve where

$$\det(H) = \lambda_1\lambda_2 = 0$$

marks a curvature transition: one principal direction changes sign.

These curves are the architectural hinges of ED. They separate:

- channel interiors (mixed curvature),
- boundary walls (positive curvature),
- horizon surfaces (negative curvature).

Curvature-Driven Reorganization
ED dynamics push curvature toward stable configurations:

- positive curvature thickens boundaries,
- negative curvature opens horizons,
- mixed curvature stabilizes channels.

This is why the architecture self-organizes even from random initial conditions.

Channels as Coherent Gradient Corridors

Alignment Condition
A channel forms where the gradient maintains coherent direction over extended distance:

$$d / ds(\nabla E / \| \nabla E \|) \approx 0$$

along a path s.

Stability Condition
Channels persist when curvature along the transverse direction is positive (confining) and along the longitudinal direction is mixed (guiding).

Mathematical Consequences

- Channels connect saddles into networks.
- Channels act as conduits for individuating flow.
- Channels define the backbone of coherent regimes.

Boundaries and Horizons as Curvature-Defined Surfaces

Boundary Condition
A boundary forms where curvature is positive in the transverse direction:

$$\lambda_{\perp} > 0$$

and gradient vectors converge.

Horizon Condition
A horizon forms where curvature is negative in the transverse direction:

$$\lambda_{\perp} < 0$$

and gradient vectors diverge.

Complementarity
Boundaries and horizons are dual surfaces emerging from the same saddle geometry. Their interplay determines whether structures stabilize, propagate, or dissolve.

Mobility Exponent and Structural Sharpness

Mobility Law
Mobility typically follows a power-law dependence on gradient magnitude:

$$M \propto \| \nabla E \|^{\alpha}$$

The exponent α controls:

- channel sharpness,
- boundary thickness,
- saddle stability,
- regime transitions.

Higher α amplifies coherent structure; lower α smooths it.

Noise Amplitude and Regime Stability

Noise Term
Noise enters as a stochastic perturbation:

$$\eta(x,t) \sim N(0,\sigma^2)$$

The amplitude σ determines whether the system remains coherent or collapses into churn.

Critical Threshold
There exists a noise threshold σc above which:

- saddles fail to stabilize,
- channels fragment,
- boundaries dissolve,
- the system enters the non-individuating regime.

Below σc, the architecture persists.

Suppression Depth and Structural Wells

Suppression Term
Suppression acts as a local sink:

$$S(E)=\beta E^{\gamma}$$

The depth and exponent determine:

- the formation of wells and pockets,
- the longevity of coherent structures,
- the balance between accumulation and dissipation.

Suppression is what allows ED to form stable cores rather than runaway growth.

Architectural Consequence: Necessary Motifs

The combined dynamics of gradients, curvature, mobility, noise, and suppression guarantee the emergence of:

- saddles,

- channels,
- boundaries,
- horizons,
- regime transitions.

These motifs are not optional—they are the mathematically necessary outcomes of the ED rule.

Regime Taxonomy in Event Density Architecture

Overview

The Event Density system exhibits four qualitatively distinct organizational regimes. Each regime corresponds to a stable configuration of gradients, curvature, and noise, and each is characterized by the presence or absence of the architectural invariants defined earlier: saddles, channels, boundaries, and horizons. The regimes are not arbitrary labels but mathematically distinct phases of the ED dynamical system.

Hyper-Coherent Regime

Mathematical Criteria

- Persistent saddles with stable eigenvalue structure:
  $\lambda_1 < 0 < \lambda_2$ with $|\lambda_i|$ large and stable over time.
- Strong gradient alignment:
  $d / ds(\nabla E / \| \nabla E \|) \approx 0$ over long distances.
- Low noise relative to curvature:
  $\sigma \ll |\lambda_i|$.
- Mobility exponent amplifies structure:
  $\alpha > \alpha_c$ for some critical value.

Geometric Signature

- Long, sharply defined channels connecting multiple saddles.
- Thick, persistent boundaries enclosing coherent pockets.
- Well-formed horizons that regulate expansion.
- A global network of individuating flow.

Structural Meaning

This regime represents the fully articulated architecture of ED. All invariants are present and stable. Structures persist for long durations and interact through well-defined channels.

Stable-Coherent Regime

Mathematical Criteria

- Saddles exist but with weaker curvature:
  $|\lambda_i|$ moderate and occasionally fluctuating.
- Channels remain aligned but may drift or bend.

- Noise is non-negligible but subcritical:
  $\sigma < \sigma_c$.
- Mobility exponent near the coherence threshold.

Geometric Signature

- Channels are present but less sharp.
- Boundaries form but may thin or deform.
- Horizons appear but fluctuate at their edges.
- Saddles persist but may migrate.

Structural Meaning
The system maintains coherent architecture but with reduced stability. Structures form, persist, and interact, but they are more sensitive to perturbations. This regime is the "working phase" of ED—organized but dynamic.

Marginal (Saddle-Dominated) Regime

Mathematical Criteria
Saddles appear but fail to stabilize:

- eigenvalues fluctuate around zero,
  $\lambda_1 \approx 0$ or $\lambda_2 \approx 0$.
  det(H) frequently crosses zero without settling.
- Gradient alignment is intermittent.
- Noise approaches the critical threshold:
  $\sigma \approx \sigma_c$.

Geometric Signature

- Channels flicker: they appear briefly and collapse.
- Boundaries form transiently but do not persist.
- Horizons open and close unpredictably.
- Saddles emerge but do not anchor long-lived structure.

Structural Meaning
This regime is dominated by the formation and dissolution of proto-structures. The architecture is visible in outline but never stabilizes. It is the transitional phase between coherence and non-individuation.

Non-Individuating Regime

Mathematical Criteria

- Noise exceeds the critical threshold:
  $\sigma > \sigma_c$.
- Curvature fails to organize:
  $|\lambda_i|$ small and rapidly fluctuating.

- Gradient sign-flip events do not persist.
- Mobility exponent too low to amplify structure.

Geometric Signature

- No persistent saddles.
- No coherent channels.
- Boundaries and horizons fail to form.
- The field remains in churn: local fluctuations without architecture.

Structural Meaning
The system cannot individuate. All architectural invariants collapse. The field behaves like noise-dominated diffusion with no stable geometric motifs.

Regime Transitions

Noise-Driven Transition
Increasing σ drives the system:

hyper-coherent → stable-coherent → marginal → non-individuating.

The critical threshold σ_c marks the collapse of saddle stability.

Mobility-Driven Transition
Decreasing the mobility exponent α reduces channel sharpness and boundary strength, pushing the system toward marginality.

Curvature-Driven Transition
Changes in suppression depth or local curvature can shift the system between stable-coherent and hyper-coherent regimes by stabilizing or destabilizing saddles.

Universality of Transitions
These transitions occur regardless of initial conditions. The phase structure is an invariant of the ED rule, not a property of specific simulations.

Regime–Invariant Relationship

Hyper-Coherent
All invariants present and stable.

Stable-Coherent
Most invariants present; stability reduced.

Marginal
Only partial invariants present; none stable.

Non-Individuating

No invariants present.

This hierarchy is the backbone of the ED phase diagram and the conceptual bridge between empirical results (ED-Arch-01) and architectural theory.

Universality of Event Density Architecture

Independence from Initial Conditions
The ED rule drives the system toward the same architectural motifs regardless of how it begins. Smooth fields, random noise, structured gradients, or even adversarial configurations all converge toward saddle–channel–boundary–horizon organization. This convergence occurs because the ED update amplifies gradient sign-flips, sharpens curvature transitions, and suppresses unstructured fluctuations. The long-term geometry depends on the rule, not the starting state.

Stability Under Parameter Deformation
The architectural structures persist across wide ranges of mobility exponent, suppression depth, and noise amplitude. Saddles remain identifiable as long as curvature retains mixed sign, channels remain coherent whenever gradient alignment is favored, and boundaries and horizons continue to form wherever curvature transitions occur. Only extreme parameter values—such as noise above the critical threshold—destroy the architecture. This robustness shows that the motifs are structural invariants of the dynamics.

Noise Robustness and Critical Thresholds
Noise perturbs the field but does not erase its architecture unless it exceeds a well-defined critical amplitude. Below this threshold, saddles stabilize, channels persist, and boundaries and horizons remain coherent. Above it, curvature cannot organize and the system enters the non-individuating regime. The existence of a noise threshold is itself an invariant: it marks a universal transition between coherent and incoherent phases.

Mobility-Law Invariance
The architectural motifs appear under a wide variety of mobility laws. Whether mobility scales linearly, superlinearly, or sublinearly with gradient magnitude, the system still produces saddles, channels, and curvature-defined surfaces. The mobility exponent controls the sharpness and persistence of these structures but not their existence. This invariance shows that ED's architecture is not tied to a specific functional form.

Curvature-Driven Necessity
The geometry of the Hessian guarantees the emergence of architectural motifs. Mixed curvature produces saddles, positive curvature produces boundaries, and negative curvature produces horizons. The curves where the determinant of the Hessian vanishes mark transitions between these regions. These curvature transitions are unavoidable in any sufficiently varied field, making the architecture a necessary geometric consequence of the ED rule.

Regime Universality
The four regimes—hyper-coherent, stable-coherent, marginal, and non-individuating—appear across all parameter sweeps and initial conditions. Each regime corresponds to a distinct configuration of curvature, noise, and mobility. The transitions between regimes are consistent and reproducible, forming a universal phase diagram. This phase structure is an invariant of the ED dynamics, not a property of specific simulations.

Dimensional Universality
The architectural motifs persist when the system is extended to higher dimensions. Saddles become saddle-sheets, channels become tubes, boundaries become walls, and horizons become surfaces. The same curvature transitions and gradient alignments appear in 3D as in 2D. This dimensional uplift shows that ED's architecture is not an artifact of low-dimensional geometry but a general feature of the rule.

Architectural Invariants
Across all conditions, the same invariants appear:

- saddle formation from gradient sign-flip
- channel alignment from coherent gradients
- boundary and horizon formation from curvature transitions
- regime transitions driven by noise and mobility
- persistence of structure under parameter deformation

These invariants define the architectural identity of ED. They are the reason the system produces the same motifs across scales, dimensions, and conditions.

Dimensional Uplift: Extending ED Architecture Beyond Two Dimensions

Persistence of Architectural Motifs Across Dimensions

The core claim is that the ED rule generates the same architectural invariants—saddles, channels, boundaries, horizons, and regime transitions—when extended from 2D to 3D and higher. The underlying reason is that these motifs arise from curvature transitions and gradient sign-flip structures that generalize naturally with dimension. The geometry becomes richer, but the architectural logic remains unchanged.

Saddles Become Saddle-Sheets

2D Structure
A saddle in 2D is defined by one compressive and one expansive principal direction.

3D Generalization
In 3D, the Hessian has three eigenvalues. A saddle-sheet forms when one eigenvalue has opposite sign from the other two:

$$\lambda_1 < 0 < \lambda_2, \lambda_3$$

or any permutation thereof.

Geometric Consequences

- The saddle becomes a *surface* rather than a point.
- This surface partitions space into regions of compression and expansion.
- Channels and boundaries attach to the sheet along its principal curvature directions.

The saddle-sheet is the organizing scaffold of 3D ED architecture, just as the saddle point is in 2D.

Channels Become Tubes

2D Structure
A channel is a narrow corridor of aligned gradient flow.

3D Generalization
In 3D, channels become tubes: coherent, elongated structures where the gradient maintains alignment along a curve embedded in three-dimensional space.

Mathematical Conditions

- Gradient alignment along a spatial curve $\gamma(s)$:

$$d / ds(\nabla E / \| \nabla E \|) \approx 0$$

- Confining curvature in two transverse directions:

$$\lambda_{\perp 1}, \lambda_{\perp 2} > 0$$

Geometric Consequences

- Tubes connect saddle-sheets into networks.
- Tubes can twist, branch, and merge in ways impossible in 2D.
- Tubes define the backbone of coherent 3D regimes.

The tube is the 3D analogue of the channel: a conduit for individuating flow.

Boundaries Become Walls

2D Structure
A boundary is a compression-dominated curve where gradient vectors converge.

3D Generalization
In 3D, boundaries become walls: 2D surfaces where curvature is positive in both transverse directions.

Mathematical Condition

$$\lambda \perp 1 > 0, \qquad \lambda \perp 2 > 0$$

Geometric Consequences

- Walls enclose coherent volumes.
- They form the “architecture” around tubes and saddle-sheets.
- They regulate the stability and persistence of 3D structures.

Walls are the structural containers of 3D ED.

Horizons Become Surfaces

2D Structure
A horizon is an expansion-dominated curve where gradient vectors diverge.

3D Generalization
In 3D, horizons become surfaces: 2D manifolds where curvature is negative in both transverse directions.

Mathematical Condition

$$\lambda_{\perp 1} < 0, \qquad \lambda_{\perp 2} < 0$$

Geometric Consequences

- Horizons mark transitions from structured interiors to unstructured exteriors.
- They regulate the flow of noise and expansion.
- They determine whether coherent volumes persist or dissolve.

Horizons in 3D are the "open surfaces" of ED architecture.

Regimes Become Volumetric Phases

2D Regimes
Regimes are defined by the stability and persistence of saddles, channels, boundaries, and horizons.

3D Generalization
The same regime structure appears in 3D, but with volumetric expression:

- Hyper-coherent: stable networks of tubes and saddle-sheets.
- Stable-coherent: persistent but deformable volumetric structures.
- Marginal: flickering tubes and transient sheets.
- Non-individuating: no stable volumetric motifs.

The phase diagram is unchanged; only the geometry becomes richer.

Curvature Transitions Generalize Naturally

The determinant-zero condition that marks curvature transitions in 2D becomes a codimension-1 surface in 3D:

$$\det(H) = 0$$

This surface separates:

- tube interiors,
- boundary walls,

- horizon surfaces,
- saddle-sheet regions.

The same architectural logic—mixed curvature inside, positive curvature confining, negative curvature expanding—applies in any dimension.

Universality Across Dimensions

The dimensional uplift demonstrates that ED's architecture is not tied to planar geometry. The same invariants appear in 3D because they arise from:

- gradient sign-flip structures,
- curvature transitions,
- mobility-driven sharpening,
- noise-driven thresholds.

These mechanisms are dimension-agnostic. The motifs change form—points become surfaces, curves become tubes—but the architectural logic remains the same.

Architectural Synthesis

A unified architectural ontology

The Event Density framework begins with a structural ontology: saddles, channels, boundaries, horizons, and regimes are the fundamental objects that constitute ED's architecture. These objects are not imposed; they arise from the geometry of the event density field itself. Saddles anchor the architecture, channels transmit individuating flow, boundaries confine structure, and horizons regulate expansion. Together, they form the minimal set of motifs required to describe coherent organization in ED.

Geometry as the engine of emergence

The mathematical formalization shows that ED's architecture is a direct consequence of gradient and curvature dynamics. Gradient sign-flips generate saddles, curvature transitions generate boundaries and horizons, and coherent gradient alignment generates channels. The determinant-zero surfaces of the Hessian act as architectural hinges, partitioning space into regions of compression, expansion, and mixed curvature. These geometric mechanisms guarantee that the ontology is not arbitrary but necessary.

Regimes as phases of architectural behavior

The regime taxonomy reveals that ED does not merely produce isolated structures but organizes itself into distinct phases. Hyper-coherent regimes exhibit fully articulated architecture; stable-coherent regimes maintain structure with moderate stability; marginal regimes flicker between order and disorder; and non-individuating regimes collapse into noise. Each regime corresponds to a characteristic configuration of curvature, noise, and mobility, forming a reproducible phase diagram. This phase structure is the dynamical expression of the ontology.

Universality as the signature of a true framework

The universality analysis demonstrates that ED's architecture is robust across initial conditions, parameter choices, noise levels, and mobility laws. The same motifs appear whether the system begins smooth or random, whether mobility is linear or nonlinear, and whether noise is low or moderate. The existence of a critical noise threshold, the persistence of curvature-driven motifs, and the invariance of regime transitions all show that ED's

architecture is not contingent—it is intrinsic to the rule. Universality elevates ED from a model to a framework.

Dimensional uplift and the scalability of structure
Extending ED to higher dimensions reveals that its architecture generalizes naturally. Saddles become saddle-sheets, channels become tubes, boundaries become walls, and horizons become surfaces. The same curvature transitions and gradient alignments appear in 3D as in 2D, and the same regime structure persists. This dimensional uplift shows that ED's architecture is not tied to planar geometry but reflects deeper structural principles that scale with dimension.

The architectural worldview
Taken together, these elements form a coherent architectural worldview:

- The ontology defines what exists.
- The mathematics explains why it must exist.
- The regimes describe how it behaves.
- The universality shows that it cannot be avoided.
- The dimensional uplift shows that it scales.

This synthesis positions ED as a general theory of architectural emergence: a framework in which structure is not imposed but arises inevitably from the interplay of gradients, curvature, mobility, and noise. The motifs observed in ED-Arch-01 are not artifacts of simulation—they are the architectural consequences of the rule itself.

Conclusion
The architectural foundations developed in this paper show that the structures observed in ED-Arch-01 are not contingent outcomes of a particular simulation but necessary consequences of the Event Density rule. By defining the ontology of saddles, channels, boundaries, horizons, and regimes, and by deriving their mathematical origins in gradient and curvature dynamics, we have shown that ED possesses a stable and reproducible architectural identity.

The regime taxonomy reveals that ED organizes itself into distinct phases, each characterized by the presence or absence of architectural invariants. The universality analysis demonstrates that these motifs persist across initial conditions, parameter choices, noise levels, and mobility laws. The dimensional uplift shows that the same architectural logic extends naturally to higher dimensions, where saddles become sheets, channels become tubes, and boundaries and horizons become surfaces.

Together, these results establish ED as a coherent architectural framework. The motifs it produces are not optional features but structural necessities. The phase transitions it exhibits are not empirical curiosities but signatures of a deeper geometric logic. The architecture is robust, scalable, and universal.

This foundation sets the stage for subsequent work: applying ED to physical, biological, and cosmological systems; exploring its predictive power; and developing a broader theory of architectural emergence across domains. The framework presented here provides the conceptual and mathematical grounding for that exploration.